MLOps Group Assignment

AG News Text Classification Pipeline

Fine-tuning DistilBERT with Docker, CI/CD, and W&B Tracking

Course:	MLOps — PGD AI Program
Institute:	IIT Jodhpur
Group Number:	Group 13
Submission Date:	June 2026

Name	Roll Number	Contribution
Ritesh Maury	G25AIT2087	GitHub Setup, Project Lead, Merges
Kosuru Yuvaraj	G25AIT2054	Training, Docker, CI/CD, HF Hub, Report
Laishram Khumanleima Chanu	G25AIT2056	Data Preparation, Inference Script
Guntakal Sandeep Kumar	G25AIT2036	W&B Tracking, Evaluation, Testing

All Submission Links

GitHub Repository:	`riteshmaury-iitj/group13-assignment-mlops`
Kaggle Notebook V1:	`kyuvarajg25ait2054/group13-ag-news-k`
Kaggle Notebook V2:	`kyuvarajg25ait2054/group13-ag-news-k`
HuggingFace Model:	`YuvarajK-g25ait2054/ag-news-distilbert`
Docker Image:	`yuvarajkg25ait2054/ag-news-classifier`
W&B Dashboard:	`g25ait2054-iit-jodhpur/mlops-assignment3`

Member Profiles & Project Links

All links are public and live at submission time.

Member Profiles

Member	GitHub	Kaggle	HuggingFace	W&B
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Project Links (all public)

Component	Link
GitHub Repository (public)	`github.com/riteshmaury-iitj/group13-assignment-mlops`
Kaggle Notebook — Version 1 (public)	`kaggle.com/code/kyuvarajg25ait2054/group13-ag-news-k`
Kaggle Notebook — Version 2 (public)	`kaggle.com/code/kyuvarajg25ait2054/group13-ag-news-k`
HuggingFace Model (public)	`huggingface.co/YuvarajK-g25ait2054/ag-news-distilbert`
Docker Image (publicly pullable)	`hub.docker.com/r/yuvarajkg25ait2054/ag-news-classifier`
W&B Project Dashboard (public)	`wandb.ai/g25ait2054-iit-jodhpur/mlops-assignment3`

1. GitHub Repository, Data Preparation and Model Selection (Tasks 1–3)

1.1 Task 1 — GitHub Repository Setup [10 marks]

Repository `riteshmaury-iitj/group13-assignment-mlops` is public with three branches: **main** (protected, PRs only), **develop** (integration), and **feature/*** (short-lived). Branch protection on `main` requires at least one approval before merging. All work was merged via Pull Request from `develop`.

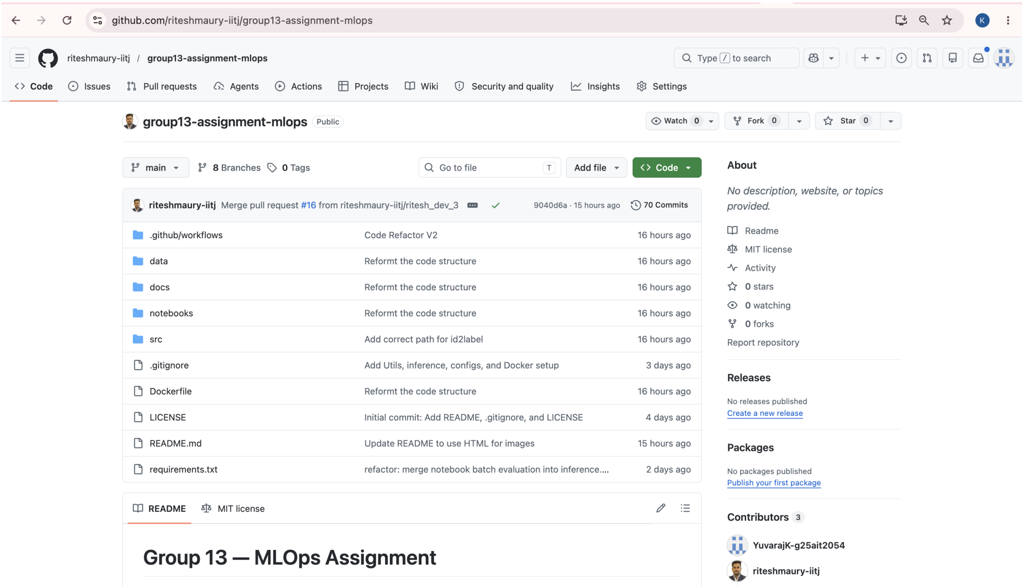

Figure 1: GitHub repository — public visibility and branch list

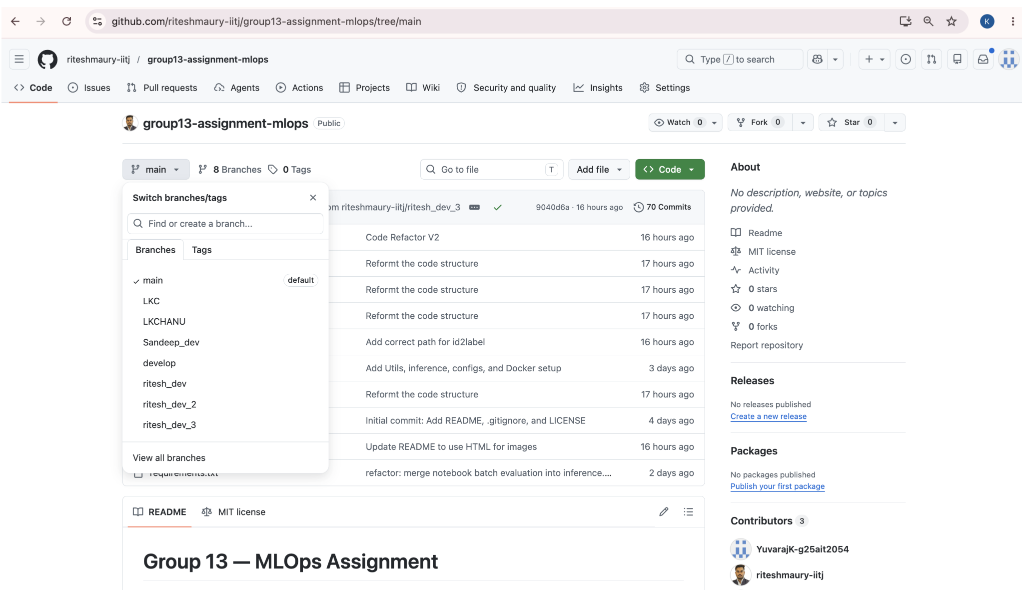

Figure 2: GitHub Settings — Branch protection rule and repo details

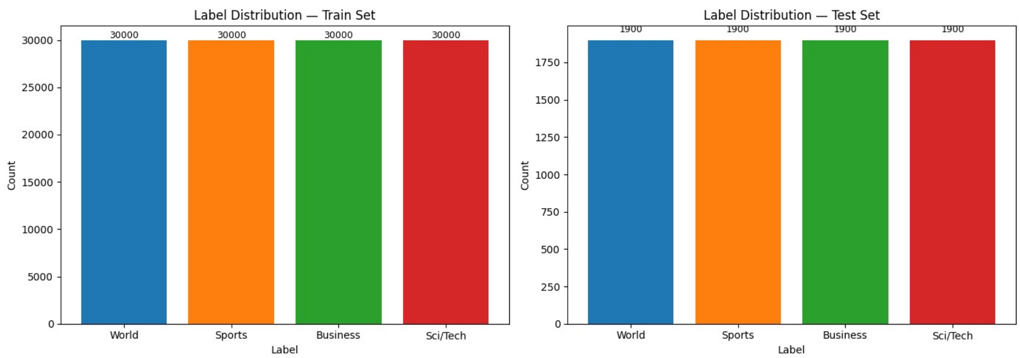

Figure 3: GitHub Settings — Collaborators: g25ait2036, g25ait2056-beep, YuvarajK-g25ait2054

1.2 Task 2 — Data Preparation & Normalisation [15 marks]

Dataset: `ag_news` from HuggingFace (120,000 train / 7,600 test, 4 balanced classes). Preprocessing applied in `prepare_data.ipynb`:

- Lowercase, remove URLs, strip HTML tags, remove punctuation, normalize whitespace.
- Label mapping saved to `id2label.json`: {`0: World, 1: Sports, 2: Business, 3: Sci/Tech`}.
- Tokenized with `max_length=128`.

screenshots/label_dist.png

Figure 4: Test set label distribution — 4 balanced classes, 1,900 samples each (7,600 total)

screenshots/length_dist.png

Figure 5: Test set text length distribution — word count per class (mean: 64 words, max: 200)

1.3 Task 3 — Model Selection [10 marks]

Selected **distilbert-base-uncased**: 66M parameters, retains ∼97% of BERT performance, and trains on AG News in ∼40 min on a T4 GPU. Loaded via `AutoModelForSequenceClassification` with `num_labels=4` and `id2label/label2id` mappings. Tokenizer uses `truncation=True`, `padding=True`, `max_length=128`.

2. Training, W&B Tracking and HuggingFace Deployment (Tasks 4–5)

2.1 Task 4 — Training on Kaggle with W&B [25 marks]

Training on Kaggle (GPU T4 ×2), tracked via W&B project `mlops-assignment3`. Secrets (W&B, HuggingFace, GitHub) stored as Kaggle Secrets, loaded via `UserSecretsClient`.

Hyperparameter Comparison:

Hyperparameter	Version 1 (run-v1)	Version 2 (run-v2)
Base Model	distilbert-base-uncased	distilbert-base-uncased
Learning Rate	2e-5	5e-5
Batch Size (train)	16	64
Epochs	3	5
Weight Decay	0.01	0.01
Max Sequence Length	128	128
Evaluation Strategy	Per epoch	Per epoch
Best Model Metric	F1 (weighted)	F1 (weighted)

Results:

Version	Accuracy	F1 (weighted)	Eval Loss
Version 1 (run-v1)	**94.34%**	**0.9434**	0.3679
Version 2 (run-v2)	94.12%	0.9412	0.3575

Version 1 achieved higher accuracy and F1; selected as the final model. Both versions exceeded 94% accuracy.

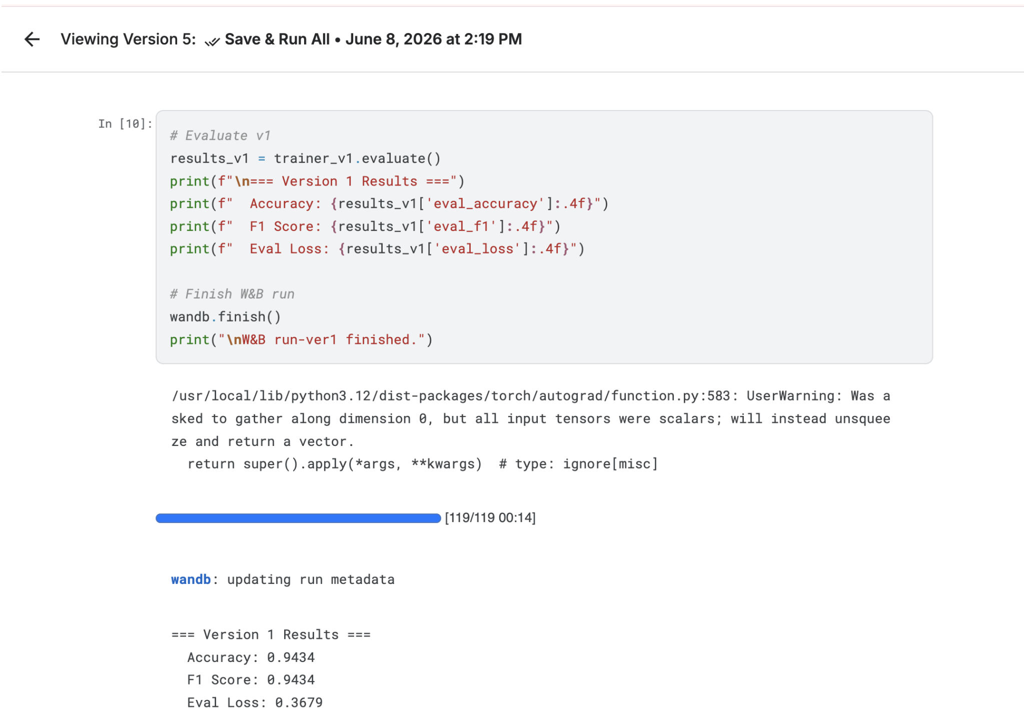

Figure 6: Kaggle notebook — Version 1 training output (eval accuracy 94.34%, F1 0.9434)

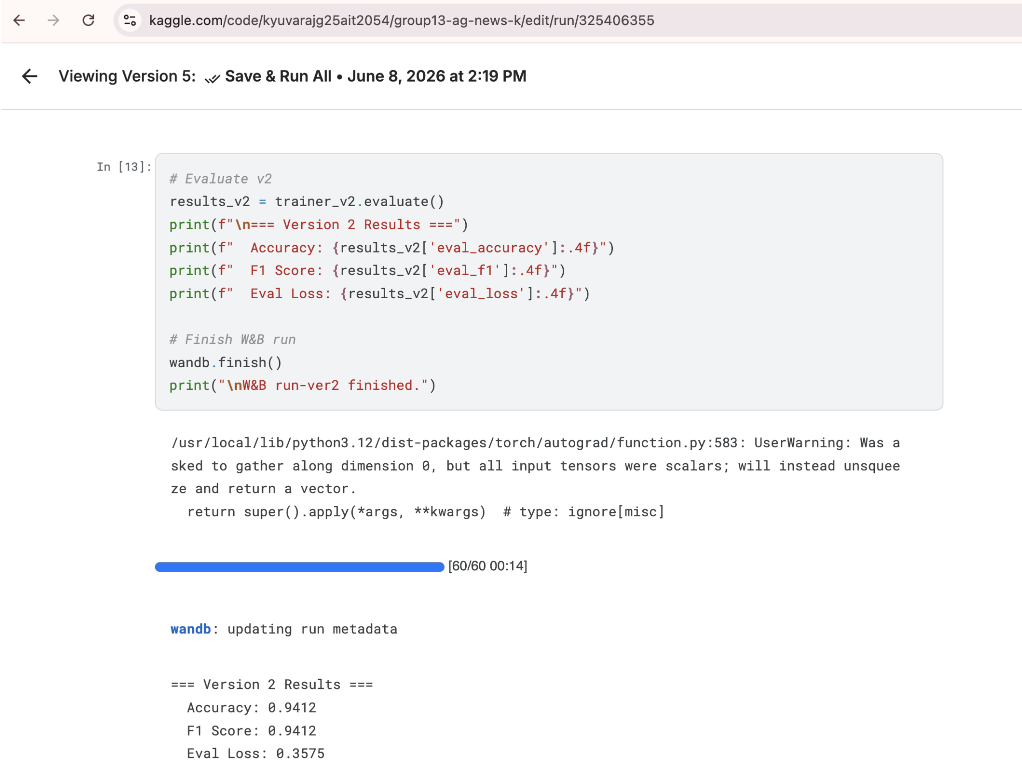

Figure 7: Kaggle notebook — Version 2 training output (eval accuracy 94.12%, F1 0.9412)

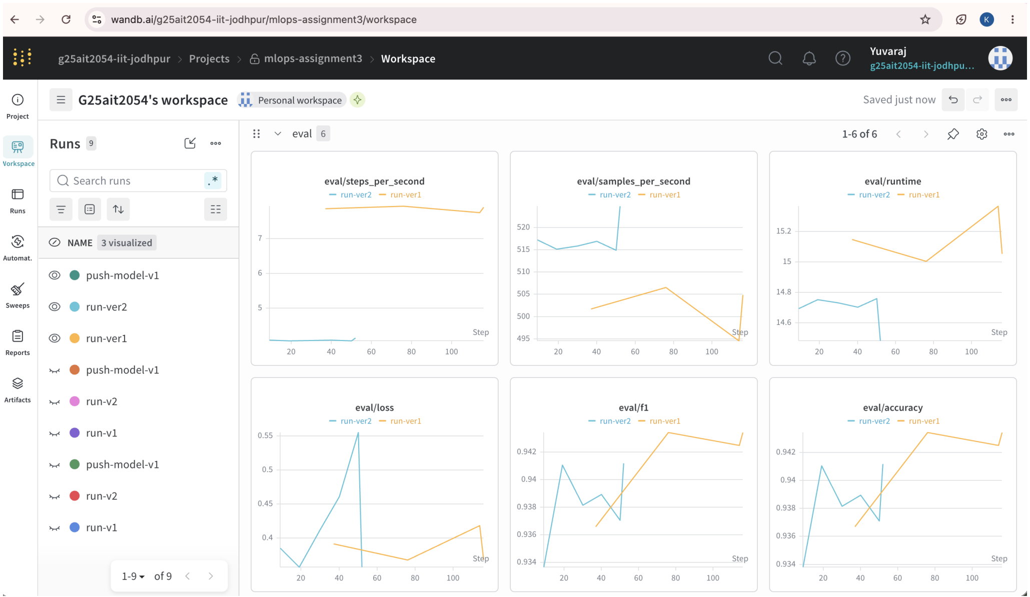

Figure 8: W&B dashboard — mlops-assignment3 showing run-v1 and run-v2 with metrics

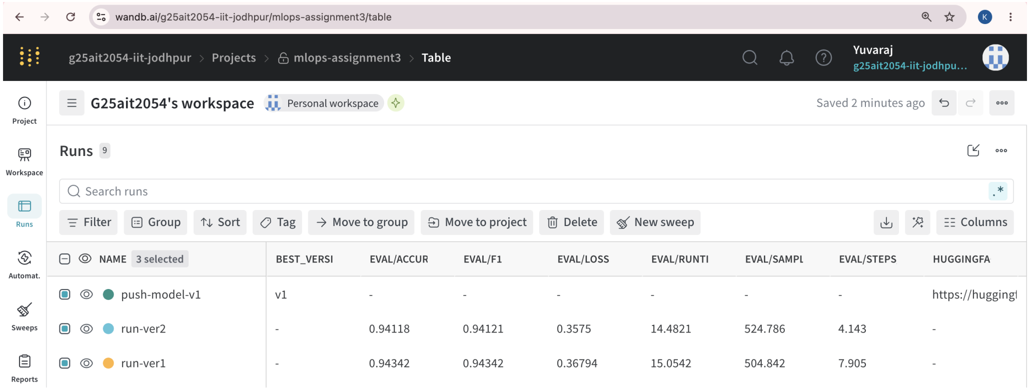

Figure 9: W&B comparison chart — eval/accuracy and eval/loss across both runs

2.2 Task 5 — Push Model to HuggingFace Hub [5 marks]

Version 1 pushed to HuggingFace Hub via `trainer.push_to_hub()`. Model is public at:

https://huggingface.co/YuvarajK-g25ait2054/ag-news-distilbert

Model card (`MODEL_CARD.md`) covers description, hyperparameters, evaluation results, and usage.

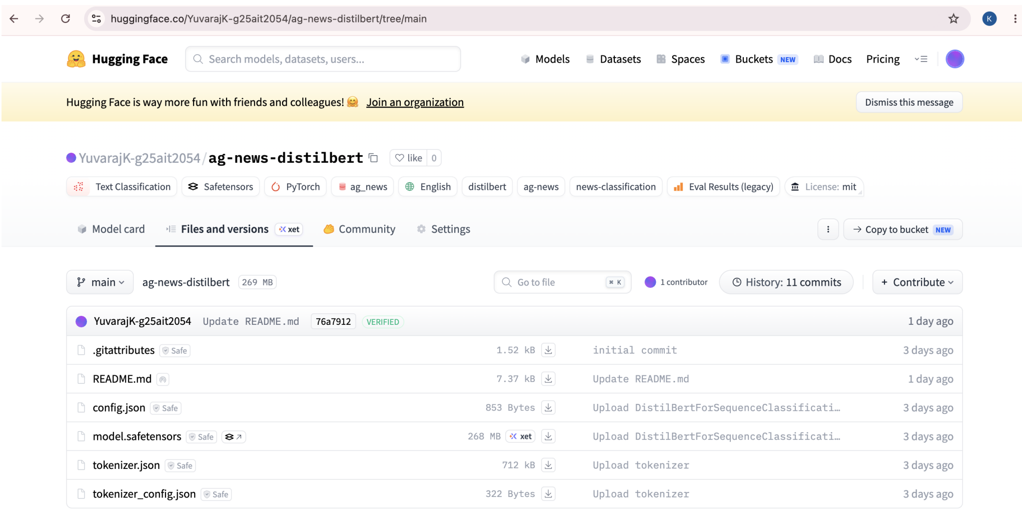

Figure 10: HuggingFace model page — YuvarajK-g25ait2054/ag-news-distilbert

3. Docker Deployment and GitHub Actions CI/CD (Tasks 6–7)

3.1 Task 6 — Dockerfile [10 marks]

Docker image hosted at `yuvarajkg25ait2054/ag-news-classifier:latest`.

- Base: `python:3.10-slim`; PyTorch `2.5.1+cpu` from PyTorch's CPU wheel index.
- Model pre-baked into image during `docker build` — no internet required at runtime.
- Default `CMD` runs `--demo` mode: classifies all 4 sample texts from `inference.ipynb` Cell 7.

```
# Demo mode (default) -- runs all 4 notebook texts:
docker run --rm yuvarajkg25ait2054/ag-news-classifier:latest

# Single text:
docker run --rm yuvarajkg25ait2054/ag-news-classifier:latest \
```

```
--text "Stock market rises on strong earnings"
```

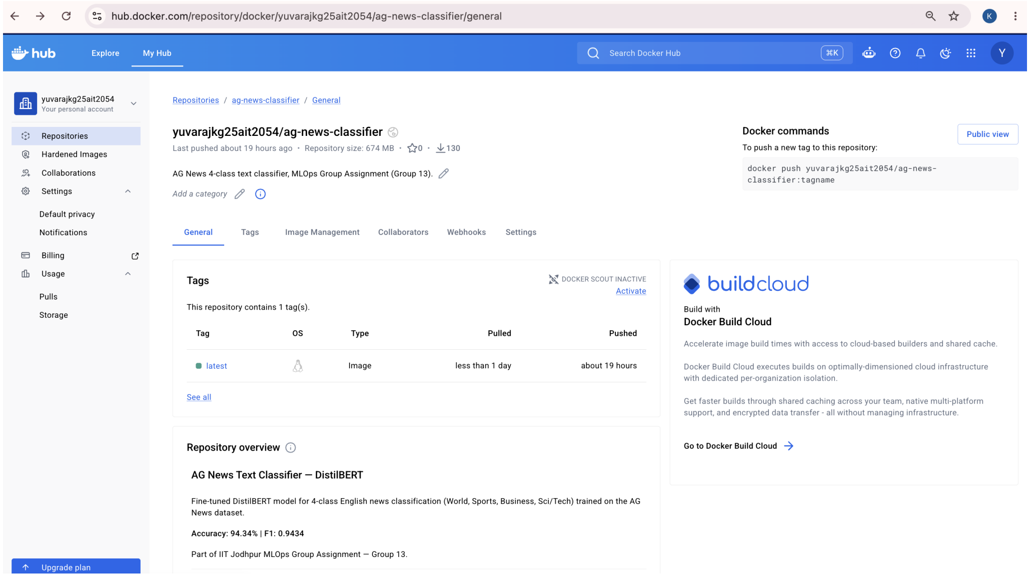

Figure 11: Docker Hub — yuvarajkg25ait2054/ag-news-classifier (publicly pullable)

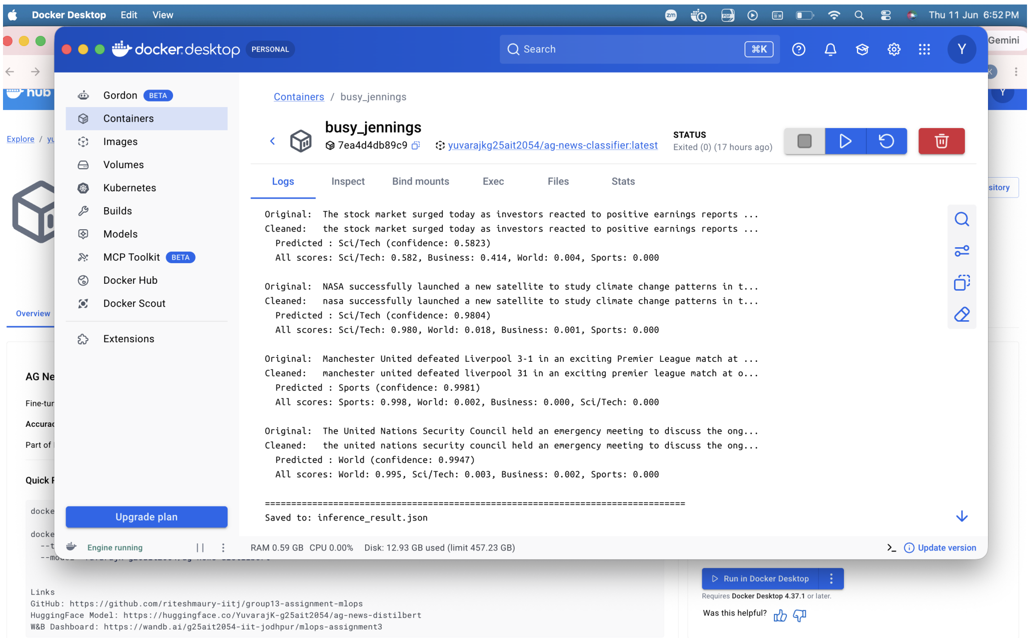

Figure 12: Docker container — demo mode classifying all 4 sample texts

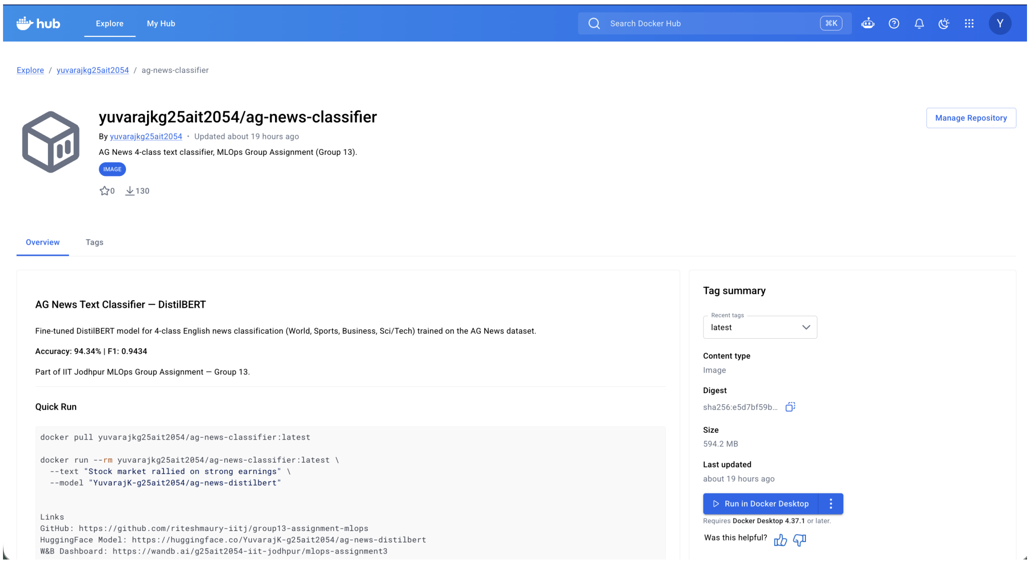

Figure 13: Docker container — terminal output showing inference results for all 4 categories

3.2 Task 7 — GitHub Actions Workflows [15 marks]

Two workflows under `.github/workflows/`:

- **ci.yml** — triggered on push/PR to `develop` and `main`. Validates required files, runs Python syntax check (`py_compile`), and validates JSON files.
- **inference.yml** — installs dependencies and runs `inference.py` end-to-end with a hard-coded test sentence against the HuggingFace model.

Both workflows pass with green checkmarks.

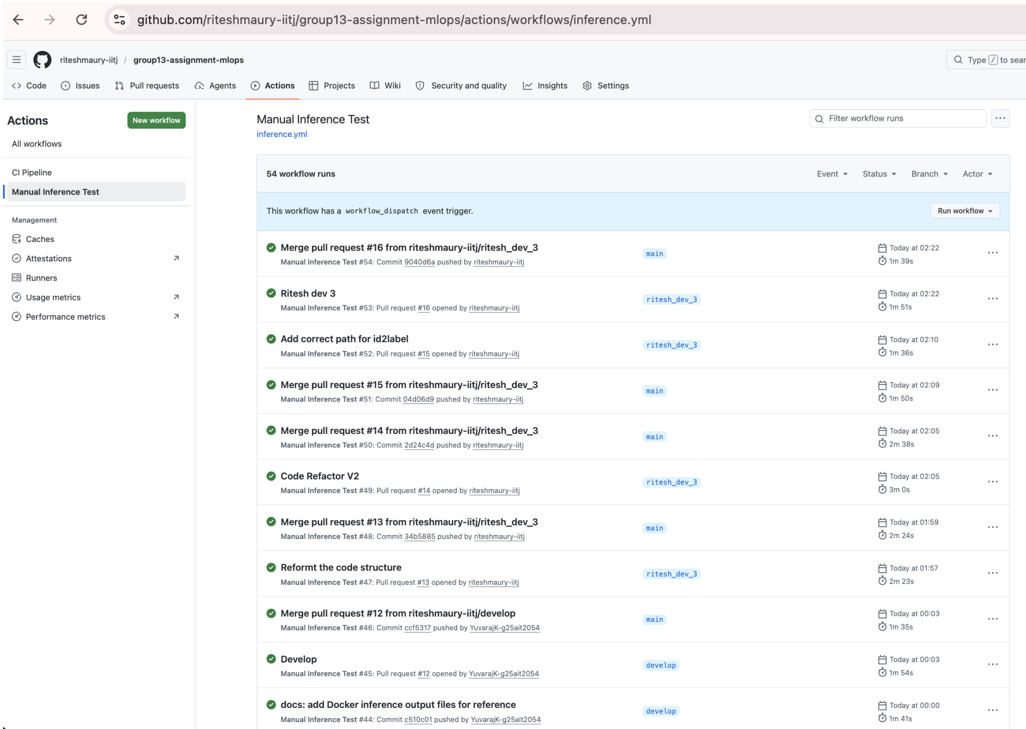

Figure 14: GitHub Actions — CI Pipeline and Inference workflow with green checkmarks

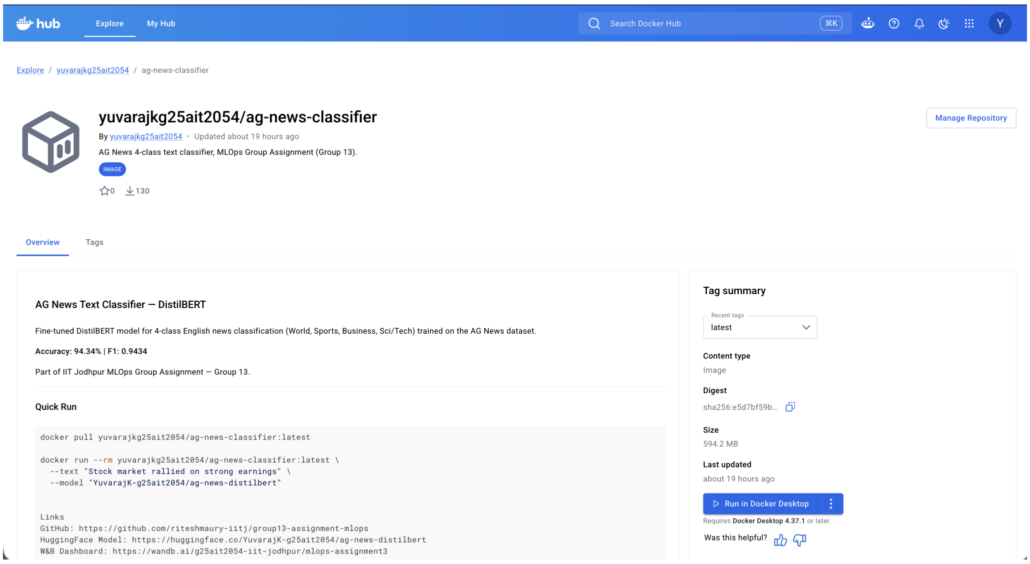

Figure 15: Inference workflow run log — classification output

4. W&B Experiment Comparison and Conclusion (Task 8)

4.1 Task 8 — W&B Experiment Comparison [5 marks]

The W&B project is publicly accessible at:

https://wandb.ai/g25ait2054-iit-jodhpur/mlops-assignment3

The project contains runs `run-v1`, `run-v2`, and `push-model-v1` (logs HuggingFace model URL to W&B summary).

- Version 1 (lr=2e-5, bs=16) converged smoothly; Version 2 (lr=5e-5, bs=64) had lower eval loss but slightly worse F1.
- Accuracy difference is small (0.22%); Version 1 selected for deployment based on higher F1.

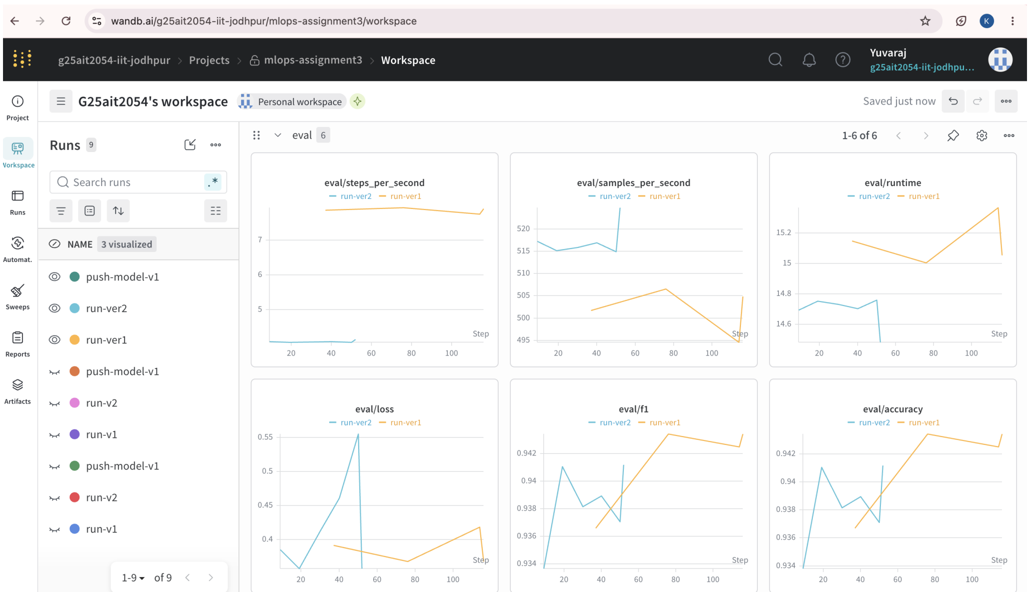

Figure 16: W&B runs table — run-v1 and run-v2 with accuracy, F1, and loss

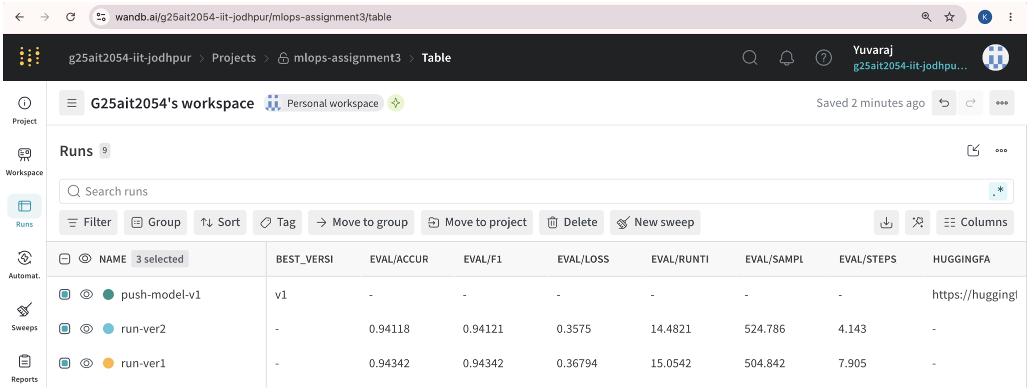

Figure 17: W&B comparison view — eval/accuracy and eval/loss with both runs overlaid

4.2 Challenges

- **Docker PyTorch:** Default PyPI `torch==2.5.1` has no CPU wheel; fixed by using PyTorch's CPU index URL (`2.5.1+cpu`).
- **DistilBERT token_type_ids:** DistilBERT does not use segment embeddings; had to remove `token_type_ids` from tokenizer output before model call.
- **GitHub Actions inputs:** `github.event.inputs` is empty on push triggers; fixed by hardcoding test values in the workflow YAML.
- **Merge conflicts:** Resolved by keeping `develop` versions of conflicted files.

4.3 Summary of Deliverables

Task	Description	Status	Link / Artefact
Task 1	GitHub Repo + Branching	Done	GitHub
Task 2	Data Prep + Normalisation	Done	`prepare_data.ipynb`
Task 3	Model Selection	Done	`distilbert-base-uncased`
Task 4	Training (v1 + v2) + W&B	Done	Kaggle
Task 5	HuggingFace Hub Push	Done	HF Hub
Task 6	Docker Image	Done	Docker Hub
Task 7	GitHub Actions (CI + Infer)	Done	Actions
Task 8	W&B Comparison	Done	W&B